Md. Yousuf **Hozaifa**

Research Assistant at AMIR Lab

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Professional Profile

AI Engineer and **Research Assistant** at **AMIR Lab** specializing in **robust ML systems** and **decentralized model training**. Developed secure **Generative AI pipelines** and **Federated Learning frameworks** using **PyTorch**, with a focus on optimizing **memory efficiency** and **model performance** for real-world deployment. Passionate about **production-grade AI solutions** and **Explainable AI (XAI)** research that balances **scalability, security, and performance** in high-impact environments.

Education

EWU(East West University)

A, 2 Jahurul Islam Ave, Dhaka 1212

B.Sc. in Computer Science and Engineering — *Feb. 2022 - Mar. 2026*

- **CGPA: 3.71 / 4.00**
- Recipient of the **Dean's List Scholarship** for academic excellence.

GSTU (Gopalganj Science and Technology University)

Gopalganj , 8100

B.Sc. in Civil Engineering — *Mar. 2019 - Dec. 2021*

- Transferred to East West University to pursue academic and research interests in Machine Learning and Computer Science.

Work Experience

Advanced Machine Intelligence Research Lab (AMIR Lab)

Dhaka, Bangladesh

Research Assistant (Dept. of NLP & Computational Linguistics) — *Feb. 2026 – Present*

- Designing and developing performance-optimized AI/ML models using **PyTorch** and **TensorFlow** focused on Advanced Machine Intelligence.
- Applying advanced research methodologies to solve real-world problems in **Generative AI**, **Large Language Models (LLMs)**, and **Explainable AI (XAI)**.
- Analyzing large-scale datasets and conducting rigorous experiments to validate novel research hypotheses.
- Collaborating with senior researchers to co-author and contribute to high-impact research papers, reports, and conference presentations.
- Participating in weekly lab meetings, technical workshops, and cross-functional team discussions to drive lab objectives.

East West University

Dhaka, Bangladesh

Undergraduate Teaching Assistant (Department of CSE) — *June 2025 – Sept. 2025*

- Served as a Teaching Assistant for **Algorithms** and **Data Communications & Networking** courses during the Summer 2025 semester.
- Assisted faculty in conducting labs and evaluating assignments on complex algorithmic problems, while mentoring students in Graph Theory, Dynamic Programming, and Network Protocols.

Udvash

Dhaka, Bangladesh

Teacher and Examiner in Mathematics — *Dec. 2019 – May 2022*

- Delivered interactive mathematics sessions for Class 12 students to clarify complex textbook concepts.
- Evaluated Engineering Admission Model Test scripts and provided constructive feedback to enhance learning outcomes.

Technical Skills

AI/ML Frameworks	TensorFlow, PyTorch, Scikit-learn
AI Security & LLMs	Latent Space Analysis, Adversarial Defense (GCG, AutoDAN), Prompt Engineering, LLaMA-3, DistilGPT2
Machine Learning	Federated Learning (FedAvg), Computer Vision (ViT, SAM), Stable Diffusion
Data Science	NumPy, Pandas, Matplotlib
Programming	Python, C++, SQL, LaTeX
Tools & DevOps	Git, GitHub Actions, Google Colab, Jupyter, Linux/Unix

Projects

The Adversarial Circuit: Causal Feature Interventions and Weight-Level Mechanisms of Prompt Injection in Agentic RAG (Active Research)

LLMs, XAI, Agentic RAG, Llama-3-8B, LangChain, FastAPI, FAISS, Latent Space Analysis

- Developing a **glass-box interpretability framework** for analyzing internal reasoning dynamics of Agentic Retrieval-Augmented Generation (RAG) systems using transformer latent-space trajectory tracking.
- Capturing **hidden-state evolution across transformer layers** during retrieval, tool invocation, and multi-step reasoning processes in autonomous AI agents.
- Investigating **latent trajectory divergence patterns** associated with hallucination formation, reasoning degradation, and indirect prompt injection attacks.
- Applying interpretability techniques including **PCA, UMAP, Tuned Lens, and Logit Lens** for layer-wise reasoning analysis and adversarial behavior detection.
- Building an experimental Agentic RAG pipeline using **Llama-3-8B, LangChain, Sentence-Transformers, and FAISS** to evaluate reasoning robustness under benign and adversarial conditions.

Tool Use Across Scale: Evaluating the Transferability of Large-Model Tool-Calling Strategies to Small Language Models *(Submitted to ICCIT 2026)*

LLMs, Tool Calling, Small Language Models, PyTorch, Hugging Face, Sentence-Transformers, API-Bank

- Developed a comprehensive evaluation framework to investigate the **transferability of large-model tool-calling strategies** to small language models (1B–3B parameters) using native function-calling interfaces.
- Constructed a benchmarking pipeline based on the **API-Bank** dataset with ambiguity-aware tool catalog generation to evaluate tool-selection performance across varying catalog sizes.
- Implemented and compared multiple inference-time interventions, including **retrieval-based tool filtering, schema simplification, and sequential tool search**, measuring their impact on tool-selection accuracy, token efficiency, and inference reliability.
- Designed a **failure-mode analysis framework** to separate retrieval-coverage failures from model-reasoning failures, enabling fine-grained diagnosis of tool-calling behavior across different model scales.
- Conducted large-scale experiments on **Llama-3.2 and Qwen2.5** models, demonstrating that retrieval-based tool filtering primarily improves **token efficiency and native tool-calling reliability** rather than raw tool-selection accuracy for small language models.

Fed-MU: Modality-Preserving Federated Unlearning for Multimodal Medical AI *(Submitted to ICCIT 2026)*

Python, PyTorch, Federated Learning, Multimodal AI, XAI, Medical AI

- Developing a **federated unlearning framework** for multimodal medical AI systems capable of selectively removing patient-specific knowledge while preserving image–text modality alignment.
- Investigating **privacy-preserving decentralized training strategies** to support compliant data deletion without requiring full model retraining.
- Designing **modality-preservation mechanisms** to reduce catastrophic forgetting and maintain diagnostic consistency after unlearning operations.
- Conducting experiments on **heterogeneous client distributions** to evaluate privacy–utility trade-offs, robustness, and representation stability in federated environments.
- Exploring **Explainable AI (XAI)** techniques for interpreting multimodal representation shifts during the unlearning process.

LEAD: Latent-Entropic Adversarial Defense for LLMs (Active Research)

LLMs, Explainable AI (XAI), PyTorch, Hugging Face, Llama-3-8B, DistilGPT2

- Designed a **dual-stage adversarial defense framework** to protect LLMs against gradient-based attacks (GCG) and fluent jailbreaks (PAIR, AutoDAN) while preserving instruction utility.
- Implemented a **low-latency entropy-based detection layer** using a DistilGPT2 proxy model, enabling efficient screening of adversarial prompts on constrained hardware (T4 GPU).
- Conducted **latent-space analysis on Llama-3-8B** to identify a consistent refusal direction separating harmful intent from benign instructions.
- **Preliminary Results:** In controlled evaluations, LEAD achieved **complete detection of gradient-based attacks (GCG, $N > 350$)** and **97.1% robustness to fluent jailbreaks (PAIR/AutoDAN)**, while maintaining a **98.86% pass rate on standard benign instructions**.
- Currently performing baseline comparisons and ablation studies to address creativity–robustness trade-offs; manuscript in preparation for conference submission.